

Hubtel API Integration Architecture & Flow

BoostUp Ghana (boostupgh.com) — Merchant UAT & Production Certification Document

Merchant Name:	BoostUp Ghana (boostupgh.com)	Date:	September 2026
APIs Used:	Online Checkout & RMSC Status API	Integration Type:	Redirect Checkout + Webhook + Requery
Production Domain:	https://boostupgh.com	Callback Endpoint:	/api/payments/hubtel/callback

1. System Architecture Overview

BoostUp Ghana integrates with Hubtel's **Online Checkout API** (Redirect flow) and **RMSC Merchant Status API**. The architecture is designed around bank-grade security, featuring server-to-server authoritative status re-verification, distributed Redis locks to eliminate race conditions, and PostgreSQL stored procedure idempotency to prevent double-crediting.

Layer / Component	Technology / Implementation	Function & Responsibilities
Frontend Client	React 18 SPA (boostupgh.com)	Deposit amount entry, checkout initiation, redirect handling, and polling fallback.
Backend API	Node.js / Next.js Serverless	PayProxy payload signing, secure UUID client references, webhook receiver.
Hubtel Gateway	PayProxy & RMSC v1 APIs	Mobile Money (MTN, Telecel, AT) and Card processing; webhook dispatch.
Database & Cache	Supabase PostgreSQL + Redis	Atomic deposit approval (RPC v2), 30s mutex lock, tamper audit logging.

2. End-to-End API Interaction Flow

The standard deposit transaction follows a 6-stage lifecycle:

Step 1: Deposit Initiation (Client -> Merchant API)

User selects Hubtel on `boostupgh.com/dashboard/deposit` and specifies amount. Frontend POSTs to `/api/payments/hubtel/initiate`.

Step 2: Hubtel Item Initiation (Merchant API -> Hubtel PayProxy)

Backend generates a cryptographic `ClientReference` (32 chars) and calls `https://payproxyapi.hubtel.com/items/initiate` using HTTP Basic Auth (`API_ID:API_KEY`). Payload includes

callbackUrl, returnUrl, cancellationUrl, amount, and clientReference.

Step 3: Hosted Modal Redirect (Hubtel -> User Browser)

Hubtel returns responseCode '0000' with checkoutUrl. The user browser is redirected to Hubtel's hosted checkout page.

Step 4: Payment Authorization (User -> Telco / Hubtel)

Customer chooses MTN Mobile Money, Telecel Cash, AT Money, or Visa/Mastercard, enters phone/card number, and authorizes the prompt on their phone.

Step 5: Asynchronous Callback Notification (Hubtel -> Merchant API)

Hubtel dispatches an HTTP POST webhook to <https://boostupgh.com/api/payments/hubtel/callback>. Our server locks the transaction in Redis (30s), verifies the payload, and makes an authoritative server-to-server call to Hubtel RMSC Status API.

Step 6: Atomic Balance Credit (Merchant API -> Database)

Once verified, `approve_deposit_transaction_universal_v2` credits user balance atomically, logs the event, and returns HTTP 200 OK to Hubtel.

3. Security & Anti-Fraud Architecture

To guarantee zero vulnerabilities against spoofing or forged callbacks, BoostUp implements a strict two-tier verification architecture:

- **Rule 1 — Zero-Trust Webhook Ingestion:** Incoming callbacks are never used to directly credit user balances. They serve purely as triggers.
- **Rule 2 — Mandatory RMSC Server-to-Server Verification:** Upon callback reception, our backend immediately calls Hubtel's RMSC Status API (<https://rmsc.hubtel.com/v1/merchantaccount/merchants/{posId}/transactions/status?clientReference={ref}>) using Merchant Basic Auth.
- **Rule 3 — Strict Amount Matching:** We compare Hubtel's verified AmountAfterFees against our database record with a 1% rounding threshold to prevent underpayment exploits.
- **Rule 4 — Distributed Redis Mutex:** A 30-second lock (smm:lock:deposit:{id}) guarantees that simultaneous webhook and status-check events cannot run concurrently.
- **Rule 5 — PostgreSQL Atomic RPC Idempotency:** Transaction status is transitioned atomically; duplicate callbacks return 200 OK without re-crediting.

4. User Return & Fallback Status Check

If a user completes payment and returns to `/payment/success?clientReference=...` before the asynchronous webhook is delivered, the frontend invokes `/api/payments/hubtel/status-check`. This endpoint queries Hubtel RMSC directly, resolves the payment status, credits the user wallet if confirmed, and displays the credited balance in real time without requiring user action.

5. Technical Endpoints Reference

Endpoint / Purpose	Method & URL
Production Domain	https://boostupgh.com
Deposit Page	https://boostupgh.com/dashboard/deposit
Hubtel Initiate Payment	POST https://payproxyapi.hubtel.com/items/initiate
Merchant Webhook Callback	POST https://boostupgh.com/api/payments/hubtel/callback
Customer Return URL	GET https://boostupgh.com/payment/success
Customer Cancellation URL	GET https://boostupgh.com/payment/cancelled
Hubtel RMSC Status Check	GET https://rmsc.hubtel.com/v1/merchantaccount/merchants/{posId}/tr ansactions/status